

Effect of Gamification on Flipped Learning in Pharmaceutical Chemistry

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Abstract—Flipped learning requires students to study and understand the lecture content before attending lectures. The lecture content was provided through instructor-generated video and lecture notes. Our earlier YouTube learning analytics showed most students attend lectures without reviewing the content material. Gamification was introduced to motivate students to study and be prepared for lectures. This study aims to assess the effect of gamification on students' engagement and motivation. The escape room was selected as the assessment tool since it requires students to answer correctly all the questions at every level. Three escape rooms, each with unique interactive features and embedded comprehension questions, were employed to facilitate understanding of fundamental chemistry concepts among first-year pharmacy students. A cross-sectional online questionnaire evaluated student engagement, and the impact of gamification on their motivation and confidence. Students' engagement was recorded using the number of views on YouTube analytics since the instructor-generated video was uploaded to YouTube. The student's number of views on instructor-generated videos increased with the escape room intervention. The study observed an increase in lecture preparedness and video engagement rates, with students reporting higher motivation, enjoyment, and confidence in their knowledge. While most students completed the escape rooms, participation rates before lectures varied, indicating the need for further exploration of barriers to completion. This gamified approach demonstrated the potential of leveraging game mechanics to transform passive learning into active problem-solving experiences, enhancing both motivation and comprehension in a flipped learning environment. Future studies could explore the longitudinal effects of gamification on student engagement.

Keywords—*Gamification, Learning Analytics, Escape Room, Motivation, Engagement*

I. INTRODUCTION

Constructivist learning theory highlights the active learning process by which individuals develop their knowledge by connecting new concepts and experiences to their existing understanding. According to Bransford and colleagues [1], this theory, which was formulated by pioneers such as Piaget, asserts that learners can either incorporate new information into their current knowledge framework or adapt that framework to

accommodate information that challenges their previous beliefs. Active learning strategies play a significant role in this process, as they encourage students to draw connections between new information and their existing mental models, thereby deepening their understanding. Educators may implement learning activities designed to address and rectify misconceptions, enabling students to reconstruct their mental models based on more accurate information. Overall, techniques that foster active engagement align with the cognitive processes deemed essential for learning within the constructivist paradigm.

The flipped classroom model was introduced to allow students to engage in active learning during class time. Essentially, "flipping the classroom" means that students first encounter new material outside of class, typically through reading or lecture videos. Then, they use class time to tackle the more challenging aspects of understanding that knowledge, such as problem-solving or discussion. According to Bloom's revised taxonomy, this approach allows students to handle lower levels of cognitive work such as gaining knowledge and comprehension outside of class while focusing on higher-order cognitive tasks like application, analysis, synthesis, and evaluation during class time. This shift provides students with the support of their peers and instructors. This model contrasts with the traditional approach, where students are first exposed to new material through lectures in class and then absorb that knowledge through homework. Hence, the term "flipped classroom."

To ensure students come to class prepared, gamification embedded with lower cognitive level assessment questions was designed to ensure students understood the lecture content after completing the instructor-generated video or lecture notes. The student's learning profile can be recorded digitally e.g. YouTube analytics providing insights into student behaviors, performance, and engagement. This data-driven approach enables the identification of patterns and trends that can inform instructional design, personalize learning experiences, and ultimately enhance student outcomes.

Escape Room was chosen as the self-assessment tool since students need to answer all questions correctly to escape. Gamification is a powerful tool to include gaming elements in a non-gaming environment. Gamification has been shown to enhance student engagement and learning outcomes [2]. By incorporating game design elements such as points, badges, leaderboards, and challenges into educational contexts, gamification transforms traditional learning experiences into interactive and enjoyable activities [3]. This approach leverages students' intrinsic motivation and curiosity, making learning more engaging and effective. It fosters a sense of achievement and competition, encouraging students to participate in their learning journey actively. Moreover, gamified learning environments can cater to diverse learning styles, promoting collaboration, critical thinking, and problem-solving skills. Gamification was shown to capture the interest of students and sustain long-term involvement to complete a task [4]. Their review highlights the diversity of gamified approaches to be tailored to various educational settings, demonstrating their adaptability to various subjects and age groups. Nonetheless, the longitudinal effect has yet to be explored [4].

Our earlier instructor-generated video on YouTube had low views before the lecture and viewing increased during study week. This learning profile was similar for two consecutive years [5] and if persists under active learning, it will hinder the learning process during the two-hour lecture. This study aims to assess the effect of gamification on the student's motivation and engagement in flipped learning.

II. METHODS

A. Participants

A cross-sectional online questionnaire was conducted for the first-year pharmacy student batch 2024/5 at Universiti Teknologi MARA, Puncak Alam, Selangor, Malaysia. As UiTM enrolled only Bumiputra with majority Malays, other demographic variables, namely gender, age, prior exposure to gamification, and types of gamifications were included. Based on the Raosoft sample size calculator, the recommended sample size for this study was 149 respondents with a population size of 242, 95% confidence interval, 5% margin of error, and 50% response distribution. The responses to this questionnaire were 163, slightly above the Raosoft sample size.

B. Study design

The topics covered in this flipped learning were from week 9 to 11 for the Fundamentals of Chemistry module consisting of lectures from Week 9 to 11 covering topics on Polyunsaturated Hydrocarbons, Alkyl halides, Substitution ($\text{S}_\text{N}1$ & $\text{S}_\text{N}2$), and Elimination ($\text{E}1$ & $\text{E}2$) reactions. One week before the lecture, the instructor-generated videos on these topics from YouTube with lecture notes were made accessible to students. The escape rooms were introduced to students to enable them to gauge their understanding of these topics before lectures. To attract the student's attention, a new escape room for each of the three weeks from Week 9 to 11 was developed.

In each of the Escape Rooms, there are a few levels/stages in which the students have to answer a few questions related to the lecture topics. After successfully answering these questions, they collect "keys/items" which help them to escape in the final stage of the game. Students who could answer all the questions correctly or managed to escape will be prompted to click on a Google Form link to claim their ranking. After the completion of the three escape rooms, the questionnaire in Google Forms was circulated online to all the first-year students. After completion, the responses were automatically recorded in the Excel datasheet. The responses were transferred to SPSS for analysis.

C. Data Analysis

The 5-point Likert scale was used to measure the student's perception of the level of difficulty on each of the questions. It was rated as strongly agree (5.0), agree, neutral, disagree, and strongly disagree (0). The descriptive mean score, standard deviation, and Cronbach's alpha value were calculated using SPSS (Version 28). Cronbach's alpha index was used to calculate the internal consistency, with values between 0 and 1, with those ≥ 0.70 being adequate and up to 0.95 being good [6]. The student's engagement with the instructor-generated video was recorded with YouTube Analytics. Pearson correlation was used to analyze the correlations between variables. The p -value of less than 0.05 is considered significant different and a Cohen's value of 0.2 was considered "small" effect size, 0.5 "medium" effect size, and an effect size of 0.8 or over was considered "large" [7].

III. RESULTS

The respondents for the questionnaire survey were 90.1% female with the remaining 9.9% male. Students range from 18-21 years old. Most students (77.8%) have experienced various gamification before entering university. They have played quizzes (96 %), flashcards (56%), puzzles (33.3%), simulation games (13.6%), strategy games (9.9%) and escape rooms (8.6%).

A. Attractive Features of Escape Room

Three different Escape Rooms were adopted from Genially (Fig 1) and were embedded with questions to evaluate the student's comprehension of the lecture contents after learning from instructor-generated video or lecture notes provided one week before the lecture. Each of the escape rooms has its storyline. The first week was the Horror Escape Room (ER1) with the sound of thunderstorms throughout the game. It has virtual interactive buttons to select answers. Students must answer questions correctly and collect inventories at all three levels of the horror house. The second week was an interactive Museum Escape Room (ER2) which was quiet without sound and only gave out sound when answered correctly. Students have to answer correctly in each room to recover all the stolen art. There are three rooms (or levels) for the student to move before recovering all the stolen art. The third Dragon Dungeon Escape Room (ER3) was an interactive four-level game to

collect four elements to kill the dragon. Throughout the escape, there is background music. Once all four elements were collected, the game proceeds to kill the dragon and the victory comes with a victorious sound. For all three escape rooms, when students answered wrongly, a sound was given out and the students had to return to the question. Students highly rated the Dragon Dungeon (ER2), followed by the Horror Escape Room (ER1) and lastly, the Museum (ER3) (Fig. 1). The ranking of their choice was due to the different features (Table 1) of each of the Escape rooms.



Fig. 1. Questions embedded in the Horror Escape room (ER1), Museum (ER2) and The Dragon Dungeon (ER3) from Genially.

The features of the Escape room (Table 1) that attracted students' attention and were rated highly were interactive (4.6 ± 0.9), audio, mysterious, and curiosity (4.5 ± 0.9). The leaderboard was rated the lowest. Cronbach's alpha of 0.91 for internal consistency indicates a strong correlation between items [8]. Although values above 0.90 may seem redundant [9], authors like van de Ven-Stevens [8] accept 0.95 as common for questionnaire scales. Leaderboard has the lowest mean (3.9 ± 1.1) as the main objective of the escape room was for the students to self-assess their comprehension on the lower level of cognitive thinking after learning new knowledge from the instructor-generated video. About 49.7% of students played in the escape rooms twice to enable them to escape. This was followed by 35.6 % of students playing three times to escape and 11.7% successfully escaped after the first trial. When they were unable to answer/escape, it meant students were unable to apply their knowledge. They refer to lecture notes/videos or sources for information and discuss with their peers. The features of the escape rooms were insignificantly correlated with the student's age or prior experience with gamification. The student's curiosity to see the ending was significantly associated with the gender with medium effect displaying a Cohen's value of 0.448 (Table 1).

TABLE 1. ATTRACTIVE FEATURES OF ESCAPE ROOM

No.	Component	Mean±stdev	Gender p-value
1	Interactive	4.6 ± 0.9	0.126
2	Audio	4.5 ± 0.9	0.042
3	Mysterious	4.5 ± 0.9	0.068
4	Curious to see the ending	4.5 ± 0.9	0.158*
5	Leaderboard	3.9 ± 1.1	0.037

Cronbach alpha value=0.91; *Significant different $p < 0.05$

The percentage of students who completed the first escape room (ER1) before the lecture was the lowest (54.6%) and increased to a consistent level (78.9%) for ER2 and ER3 (Fig. 2). The combined percentage of student completion increased from ER1 (84.2%) to ER2 (90.5%) but reduced to ER3 (79.3%). Thus, for every lecture, only 79% of students were prepared. The remaining 21% did not complete the Escape rooms before the lecture and cited a lack of time to complete them. The accessibility of Escape rooms after the lecture allows them the convenience to complete or revise. Students have different learning traits and online interactive escape rooms allow accessibility according to the student's time making it personalized learning [10].

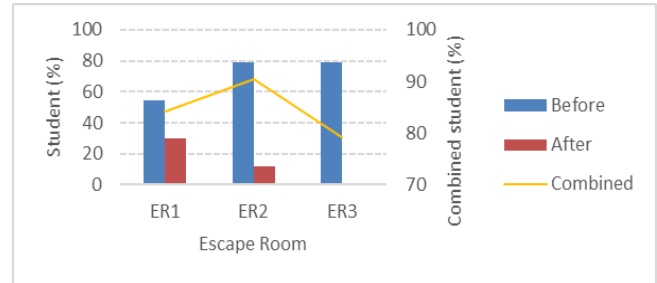


Fig. 2. Percentage of escape room completion

B. Quality of questions and Students' Emotion

The students found the questions in the Escape Room relevant to the topics they studied online (Table 2) in flipped learning, and they enjoyed testing their comprehension of the new knowledge they acquired. The assessment questions embedded in the ER were enjoyable when they could answer them, which motivated them to study. The questions were direct but required them to relate to the theories learned, thus making the escape room challenging.

TABLE 2. STUDENTS EMOTION AND QUALITY OF QUESTIONS

No.	Description	Mean ± stdev	Gender p-value	Prior experience p-value
1	I find the questions challenging.	3.9 ± 1.0	0.124	0.001
2	I find the questions relevant to the topics.	4.6 ± 0.9	0.098	0.098
3	I find the questions direct	4.1 ± 0.9	0.138*	0.217*
4	I need to relate the theories to the answer.	4.3 ± 0.9	0.144*	0.133
5	I enjoyed testing my comprehension with Escape Room.	4.6 ± 0.8	0.095	0.089
6	It motivates me to study relevant topics.	4.5 ± 0.8	0.145*	0.126
7	The questions in the Escape Room make me more confident with the topic.	4.4 ± 0.9	0.091	0.162
8	After completing the escape room, I search online or books regarding the completed topic.	3.8 ± 1.1	0.021	0.077

Cronbach alpha value=0.93; *Significant different $p < 0.05$

Being able to "escape" or answer all the questions correctly makes them more confident. Some students were resourceful and searched online or in books after completing the ER. Gender and prior experience with gamification were significantly correlated to direct questions (Table 2) with a medium effect of Cohen's value of 0.392 and 0.531, respectively. Questions requiring students to relate to theories and motivation to study relevant topics (Table 2) were significantly correlated to gender with a medium effect of Cohen's value of 0.407 and 0.411, respectively. The student's age was insignificant.

Although a small percentage (8.6%) of students have experience with the escape room before entering university, a majority (89.6 %) of students prefer the Escape Room to self-assess their comprehension. Some students commented, "The flipped sessions are engaging and keep me from feeling bored or sleepy before class. I hope this teaching method can continue in the future", "I love the games", "Make more games", "Escape room is really fun to play, and it helps me to understand", etc.

C. Learning Analytics from YouTube

The video uploaded to YouTube generated the number of views (Fig. 3). The spike in the number of views was increased for the same lecture topic repeated twice to two different groups of students. The escape room has increased the number of views during the lecture week compared to the revision week (Fig. 3). This learning profile was reversed in earlier studies without the escape room [5].

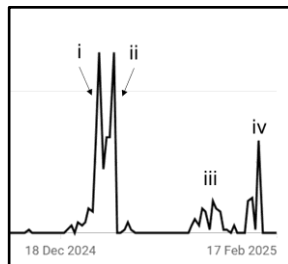


Fig. 3. Number of views from YouTube on the same lecture topic i) first lecture; ii) second lecture; iii) revision week & iv) exam week.

IV. DISCUSSIONS

Our earlier Learning Analytics from YouTube have shown a low view ratio of instructor-generated video lectures on YouTube one week before the lecture [5]. To ensure students read before attending lectures, escape rooms were introduced.

The design of the Escape room is important to attract the student's participation to complete all the questions. The features or elements of the game mechanics of the Escape room that captured the students' attention were interactive, audio, mysterious, curiosity, and leaderboard (Table 1). These features increase the learning process's sense of fun and challenge [11]. The leaderboard was rated the lowest (3.9 ± 1.1) effect on students because the purpose of the escape room was for self-assessment. Curiosity to know the ending was the main attractive feature of the escape games which was

significantly associated with student's gender with medium effect. The Escape Room provided an excellent tool to enhance their intrinsic motivation factor, which was significantly associated with gender with medium effect, as the competitors in the Escape Room were themselves. Most students (97%) play the Escape Room between 1 to 3 times to escape. The priority of this Escape Room was to challenge themselves rather than amongst their peers. Thus, in this case, the leaderboard for the escape room is less profound among the students. Careful customization of the type of game to the specific educational context to meet the needs of all learners is required [12].

In the second part of the game, we look into learning mechanics. Since the escape room was designed for students to self-evaluate their understanding of flipped learning, it has to be encouraging with simple but critical questions. It is of high priority that all the students complete this self-assessment or be able to escape the room before attending the lecture. The designed questions achieved their objective as the students found the questions relevant and required them to relate to the theories but were not too challenging (Table 2). They were able to escape the rooms. The design has increased their motivation (Table 2) and engagement in the ratio of video view (Fig. 3) to search for answers. In the comments on the questionnaire, most students enjoyed the interactive game and being able to apply the knowledge they learned from the video to answer the questions embedded in the Escape room. They also wanted more escape rooms. Prior experience with gamification does not correlate with the game features but was significantly associated with a medium effect on the direct questions embedded in the escape rooms.

At present, mechanical checking on the number of students who have completed the escape room before and after the lecture was required to encourage students to fully participate. Thus, for the first lecture, the percentage of participants between groups was presented and those completed will be shown next to their online attendance. Thus, the number of students who completed the lecture increased for the second escape room (ER2). Though mechanical checking is done and presented during each lecture, the highest percentage of participation in the escape room before lectures was 79%. Most students completed the escape room 12 hours before the lecture. Those who did not complete cited a lack of time and some commented that it was preferable to do the escape room in their leisure time as they felt pressured to complete it at a fixed time. Since the escape rooms and learning materials were accessible online, these students could tailor the time according to their schedule making it personalized learning.

The three escape rooms used must provide different game aesthetics since student enrollment was high and each student has their learning preference. Aesthetics, in the context of game design, refers to the player experience. Players experience aesthetics first and then immerse themselves in the game flow (dynamics, mechanics) [13]. The students highly rated the aesthetics of the dragon dungeon, followed by the horror room and museum. Both the dragon dungeon and horror room have

audio throughout the game. The audio has influenced the students' preferences. The museum escape room was quiet throughout the game and gave out sound when answered correctly. Some students have noted they can remember better with the escape room. Most students liked the escape room as they enjoyed testing their comprehension with it (Table 2). It motivated them to study and were confident with the knowledge they learned (Table 2). This was observed in the increased view of YouTube analytics (Fig. 3). As educational environments become increasingly digital, the role of learning analytics offered powerful tools to foster more effective and meaningful learning experiences.

Gamification leverages interactivity in comprehension questions, transforming passive learning into an active problem-solving adventure. This captivates learners' attention and fosters deeper engagement with fundamental chemistry. Students experienced increased motivation and confidence as they successfully progressed in each level of the game. Immediate feedback is a staple in gaming allowing students to learn or correct their theoretical concepts. The limitation of this study is that the gamification effect is studied in the first-year students and the longitudinal effect was not assessed.

V. CONCLUSION

The integration of gamified escape rooms into flipped classroom learning demonstrates significant potential to enhance student motivation, engagement, and understanding. This study highlights how game design elements, such as interactivity, thematic audio, and immersive challenges, create an engaging learning environment that aligns with the constructivist paradigm. Students reported improved confidence in their knowledge and enjoyed testing their comprehension, suggesting that gamification can make learning more effective and enjoyable. The increase in video view ratios and consistent completion of escape rooms underscores the effectiveness of gamification in fostering active learning behaviors. However, variability in escape room participation before lectures suggests the need to address individual preferences and external factors influencing engagement. Incorporating a broader range of game aesthetics and adaptive mechanics may further cater to diverse learning styles. Additionally, understanding and mitigating barriers to participation could improve overall effectiveness. Future research could explore long-term impacts on learning outcomes and the scalability of gamified flipped learning models. Overall, this study supports the use of gamification as a powerful tool to enhance the flipped classroom experience, making learning more interactive, inclusive, and aligned with 21st-century educational goals.

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